GROUP PROJECT CONTRIBUTION STATEMENT

Course: Predictive Analytics • Cakrawala University
Academic Year 2026/2027

Project Title:	Hyperscaler Capital Expenditure and Cloud Segment Revenue Growth: An Empirical Time-Series & Econometric Analysis
Project Theme:	Finance and Business Analytics
Task Type:	Time-Series Forecasting (One-Step-Ahead Walk-Forward Validation)
Course Registry:	s.id/listdatapred02
Submission Date:	19 October 2026 (20:00 WIB via RISE)

1. Group Membership & Shared Responsibility

This project was planned, executed, and documented collaboratively. The analytical, technical coding, interpretation, and presentation responsibilities were distributed as follows: TITA NOVIANA (40%), SUCI FRANSISCA SISILIA R (40%), and NADIRA DIKA KAIVALYA (20%). Each member has developed a comprehensive, end-to-end understanding of the entire analytical workflow and is fully prepared to explain and defend the methodology, implementation choices, and findings during the technical interview.

Full Name	Student ID	Primary Assigned Role	Share
TITA NOVIANA	24120500011	Lead Data Engineering & Econometric Modeling	40%
SUCI FRANSISCA SISILIA R	24120500008	Lead Machine Learning & Hyperparameter Tuning	40%
NADIRA DIKA KAIVALYA	24110500023	Lead Business Translation, Regime Analysis & Governance	20%

2. Detailed Allocation of Analytical, Coding, Interpretation & Presentation Roles

2.1 TITA NOVIANA (Student ID: 24120500011) — Role: Lead Data Engineering & Econometrics

- **Analytical:** Formulated core predictive questions (H1: capex transmission, H2: ML vs. baseline); established one-step-ahead expanding-window walk-forward validation logic; conducted Augmented Dickey-Fuller (ADF) stationarity tests and Cross-Correlation Function (CCF) lag diagnostics.
- **Coding:** Built automated ingestion pipelines for SEC EDGAR XBRL companyfacts API and FRED Federal Reserve macroeconomic series; engineered chronological lag structure (lags 1-4) with strict zero-leakage controls; programmed Naive Persistence Baseline and AR(1) ARIMA maximum-likelihood models.
- **Interpretation:** Evaluated baseline persistence dynamics versus autoregressive models; analyzed downward life-cycle drift in hyperscaler growth rates; quantified sample size constraints across quarterly reporting windows.
- **Presentation:** Delivered Pitch Sections 1 & 2 (Problem Context, CFO Capital Allocation Dilemma, & Zero-Leakage Architecture); prepared for Technical Interview on data auditing and stationarity.

2.2 SUCI FRANSISCA SISILIA R (Student ID: 24120500008) — Role: Lead Machine Learning & Tuning

- **Analytical:** Designed systematic hyperparameter search spaces for the two strongest non-baseline model families (Ridge & XGBoost); established expanding-window TimeSeriesSplit protocols to prevent temporal contamination; evaluated shrinkage mechanics in ill-conditioned financial panels.
- **Coding:** Implemented regularized Ridge (L2) and Lasso (L1) models with dynamic path optimization; built shallow-tree constrained XGBoost Regressor (max_depth=2-3); generated master evaluation tables (RMSE, MAE, Directional Accuracy) and diagnostic visual residual plots.
- **Interpretation:** Executed the Default-versus-Tuned comparative performance audit required by the rubric; interpreted Lasso coefficient shrinkage to zero (0.000) for all lagged capex features; synthesized empirical evidence disproving multi-quarter capex predictive superiority.
- **Presentation:** Delivered Pitch Section 3 (Candidate Model Benchmark, Default vs. Tuned Trade-offs, & Non-Causal Feature Coefficients); prepared for Technical Interview questions on regularized ML and hyperparameter tuning.

2.3 NADIRA DIKA KAIVALYA (Student ID: 24110500023) — Role: Lead Business Translation & Regime Analysis

- **Analytical:** Formulated the structural regime shift test (H3: pre-2023 cloud maturation vs. post-2023 Generative AI boom); audited Microsoft FY2024 segment reclassification (-17% revenue recast); designed the transparent 2030 compounding scenario exercise.
- **Coding:** Curated, cross-checked, and verified 121 quarterly cloud segment revenue disclosures directly from primary Form 8-K Exhibit 99.1 earnings releases; developed pre- vs. post-2023 regime evaluation routines; generated the 2030 fan chart visualization with widening uncertainty bands.
- **Interpretation:** Formulated the 3 Core Evidence-Based Findings and 2 Feasible Strategic Recommendations for Chief Financial Officers; analyzed server depreciation overhang (4-6 year schedules), internal AI R&D; absorption, and margin compression risks.
- **Presentation:** Delivered Pitch Section 4 (Strategic C-Level Actionable Recommendations, Capital Budget Decoupling, & 2030 Scenarios); led group defense during stakeholder panel Q&A; and technical interview.

3. Academic Integrity and Individual Ownership Affirmation

We hereby affirm that: (1) The submitted predictive analysis represents our original, collaborative work conducted specifically for the Predictive Analytics course at Cakrawala University; (2) All financial datasets and macroeconomic indicators are authentic, public, and verifiable against primary SEC EDGAR and Federal Reserve sources; and (3) The individual ownership is allocated as TITA NOVIANA (40%), SUCI FRANSISCA SISILIA R (40%), and NADIRA DIKA KAIVALYA (20%), with each member ready to independently explain and defend any part of the analytical report.

TITA NOVIANA	**SUCI FRANSISCA SISILIA R**	**NADIRA DIKA KAIVALYA**
Student ID: 24120500011	Student ID: 24120500008	Student ID: 24110500023
Date: 19 October 2026	Date: 19 October 2026	Date: 19 October 2026